

too far to relay two-way communication with acceptable propagation delays. Propagation delay in a geosynchronous orbit is about forty times longer than in an LEO.

The satellites take up their positions on the lower orbits covering 1200-1800 kilometres in diameter. It is estimated that four to five satellites are enough to ensure broadband at one point of time all across the Indian geographical region.

Packed with a bandwidth of 100Gbps per satellite, the users on Earth can have up to 50Mbps and 400Mbps for business users. The beauty of it is that, this speed does not depend on the geographical location of the user. The speed at which the Internet streams for people accessing from a crowded city or a remote hamlet in the Himalayas remains the same.

The vision we witness here is similar to the telecommunication revolution where people linked up to the mobile-phone technology and discarded the old immediate telephones. In time this technology is capable of providing a major leap of accomplishment for the Internet connectivity.

The ground nodes would be set up on strategic locations on the ground, though most of the infrastructure would be floating in space. The ground nodes would connect and talk to servers that are located on the Earth's surface.

HOW ASTROME FARES IN SPACE COMPETITION

Parameter	OneWeb	Viasat-3	Astrome
Frequency	Ku band	Ka band	mm-wave
Orbit type	LEO	GEO	LEO
Capacity per satellite	8Gbps over 1000km diameter	1Tbps over 7500km diameter (or 17Gbps over 1000km diameter)	100Gbps over 1000km diameter
Number of satellites	648	3	100
Targeted geographical regions	Global	Global	South Asia and South-East Asia
Total capacity	5.184Tbps	3Tbps	10Tbps

Other competing ideas

Providing the Internet from air or space is of much interest these days. But Astrome is a little different from the other players in this space. Google's 'Project Loon' plans to make the Internet available to remote locations via balloons moving in the jet stream at 18km altitude from ground. Facebook has plans to beam broadband signals from drones and satellites. Another player, SpaceX, plans to build a network of four thousand satellites in LEO by 2030.

There is also OneWeb, backed by Bharti Enterprises in India, who would be placing 648 satellites in LEO by 2018. Viasat is planning to provide the Internet service with up to 1Tbps by launching three GEO satellites (Viasat-3).

"Our approach to the problem is quite different from the other players like OneWeb and ViaSat," says Dr Prasad Bhat (see Table). What differentiates Astrome from OneWeb

and ViaSat is that Astrome operates in the millimetre wave region of the electromagnetic spectrum, which is between 30GHz and 300GHz. Millimetre waves have very narrow transmission beam width and high bandwidth band allocations. This allows it to have significantly higher capacity per satellite.

Future plans

Astrome wants to involve individuals and organisations to build a strong team and become a collaborative effort. Their talks with ISRO for launching the satellites are already on. By 2018, Astrome, hand in hand with ISRO, plans to take help of experienced satellite design and manufacturing companies to create a satellite assembly line.

Astrome also plans to actively engage with the Internet of Things (IoT) industry, which is expected to significantly influence the technology landscape in the next five years. It intends to provide their IoT devices with location-free, reliable, high-bandwidth Internet from as early as 2019.

The Indian space programme shows a road map for science and technology to improve the quality of life of the masses. At Astrome, it is considered a coincidence that, together with ISRO, an ambitious idea like this is taking shape to use space technology for providing location-free, high-speed Internet. Astrome is being incubated at IISc, a place where the foundations for the Indian space programme were laid way back in the 1950s. **EFY**

About the innovators



Dr Neha Satak

"Our motto is 'Connectivity anytime, anywhere,'" says Dr Neha Satak and Dr Prasad Bhat, serial entrepreneurs and co-founders of Astrome.

With a PhD in aerospace engineering from the Texas A&M University, Neha is well versed with satellite technologies. Prior to starting Astrome, she worked on asteroid deflection technologies and started Scientific Preparatory Academy for Cosmic Explorers (SPACE), a non-profit organisation aiming to promote hands-on space education around the world.

Prasad, who has a PhD in computer science from the Indian Institute of Science, also brings in a lot of entrepreneurial experience. Before joining hands with Neha to start Astrome, Prasad co-founded Streamold Technologies, a company that builds exciting products around visual technology.

They have also roped in technology advisors from ISRO, IISc, IISc and hi-tech communication industry in India, and business advisors from the Silicon Valley, United States.



Dr Prasad Bhat